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 Studierichting: Informatica
 Oefeningenreeks 4A nr. 57.3

$$\int \frac{dx}{1 + \tan x}$$

$$t = \tan x$$

$$dt = (1 + t^2)dx$$

$$\frac{dt}{1 + t^2} = dx$$

$$\Rightarrow \int \frac{dx}{1 + \tan x} = \int \frac{1}{1 + t} \cdot \frac{1}{1 + t^2} dt$$

$$= \int \frac{1}{(1 + t)(1 + t^2)} dt$$

$$\frac{1}{(1 + t)(1 + t^2)} = \frac{A}{1 + t} + \frac{Bt + C}{1 + t^2}$$

$$A = \frac{1}{1 + t^2} \Big|_{t=-1} = \frac{1}{2}$$

$$Bi + C = \frac{1}{1 + t} \Big|_{t=i} = \frac{1}{1 + i}$$

$$Bi + C = \frac{1 - i}{(1 + i)(1 - i)} = \frac{1 - i}{2}$$

$$Bi + C = \frac{1}{2} - \frac{1}{2}i$$

$$\Rightarrow (B, C) = \left(-\frac{1}{2}, \frac{1}{2}\right)$$

$$\Rightarrow \int \frac{1}{(1 + t)(1 + t^2)} dt = \int \left(\frac{\frac{1}{2}}{1 + t} + \frac{-\frac{1}{2}t + \frac{1}{2}}{1 + t^2} \right) dt$$

$$= \frac{1}{2} \int \frac{1}{1 + t} dt - \frac{1}{2} \int \frac{t}{1 + t^2} dt + \frac{1}{2} \int \frac{1}{1 + t^2} dt$$

$$= \frac{\ln|1 + t|}{2} - \frac{1}{2} \int \frac{t}{1 + t^2} dt + \frac{\text{Bgtan } t}{2} + C$$

$$u = 1 + t^2$$

$$du = 2t dt$$

$$\frac{du}{2} = t dt$$

$$\Rightarrow \int \frac{t}{1 + t^2} dt = \frac{1}{2} \int \frac{1}{u} du$$

$$\begin{aligned}
&= \frac{1}{2} \ln |u| \\
&= \frac{1}{2} \ln |1 + t^2| \\
&\Rightarrow \int \frac{dx}{1 + \tan x} = \frac{\ln |1 + t|}{2} - \frac{1}{4} \ln |1 + t^2| + \frac{\operatorname{Bgtan} t}{2} + C \\
&= \frac{\ln |1 + \tan x|}{2} - \frac{1}{4} \ln |1 + \tan^2 x| + \frac{x}{2} + C
\end{aligned}$$

References